

Atlantoaxial (AA) Luxation

What is it?

Atlantoaxial luxation is instability between the **first two bones of the neck** (the atlas and axis). When these bones shift, they can press on the spinal cord and cause pain or serious injury.

Which dogs are affected?

Mostly **toy and small breeds** — Yorkshire Terriers, Chihuahuas, Toy Poodles, Pomeranians, Miniature Dachshunds — usually **under 2 years old**, though any age or breed can be affected.

Causes

- **Congenital:** abnormal development of the *dens* (a peg of bone that normally anchors the two vertebrae)
- **Traumatic:** rough play, falls or accidents

Signs

- Neck pain and stiffness; head held low
- Reluctance to move the head and neck
- Weakness or wobbliness in the legs
- Severe cases: paralysis or trouble breathing

Diagnosis

Neurological exam, neck **X-rays** (often under sedation for careful positioning), and **CT or MRI** for detail.

Treatment

Conservative management — for mild cases or when surgery is not possible: strict confinement, a neck

brace/splint and pain medication. The joint remains unstable, so improvement may be temporary.

Surgery — often recommended. The vertebrae are stabilized with pins, screws, wires or bone cement so they fuse permanently. Success is good, especially when done before severe spinal cord damage.

After treatment

- Strict rest and limited activity for several weeks
- **Harness instead of a collar — for life**
- Recheck visits; physical rehabilitation in some cases

Prognosis

Depends on how injured the spinal cord was before treatment, the success of stabilization and following the aftercare plan. Many dogs recover well, though some signs may persist in severe cases.

Prevention for at-risk breeds

- Always use a harness, never a neck collar
- Prevent jumping from furniture; use ramps
- Support the head and neck when holding puppies

Seek emergency care for

Sudden neck pain, inability to walk, worsening weakness or **difficulty breathing**. AA luxation can be a true emergency.